PYTHON ACTIVITY-BASED LEARNING

Case Study 09: Integrated Student Record Consistency Checker

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Student Name: _____ **Roll No:** _____

Branch / Section: _____

Problem Statement

Validate student records stored as tuples and detect duplicate IDs, missing names, invalid marks and attendance.

Objectives

• Validate identifiers and names.

• Check marks and attendance ranges.

• Detect duplicate student IDs.

Assumptions

• Each tuple contains ID, name, marks and attendance.

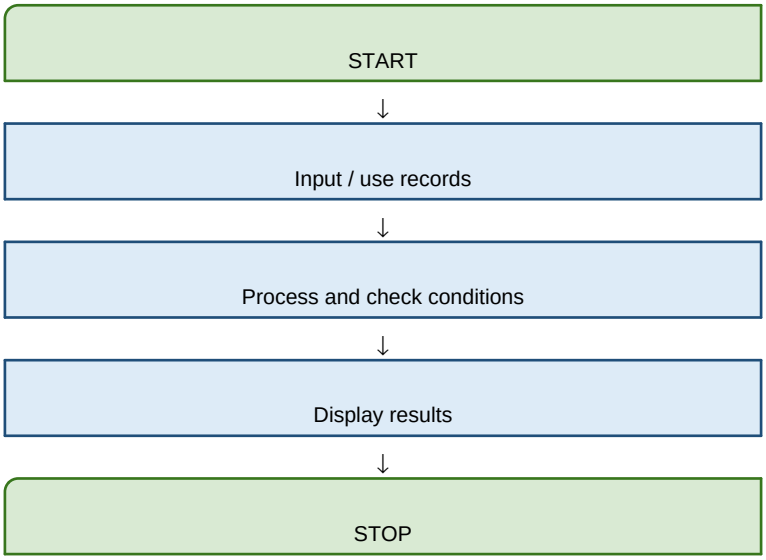
• Marks and attendance must be between 0 and 100 inclusive.

• A blank name or repeated ID is invalid.

Algorithm

1. Store student records as tuples.
2. Create an empty list for IDs already checked.
3. Unpack each record.
4. Check duplicate ID, blank name, marks range and attendance range.
5. Display valid or invalid status with reasons.

Flowchart



Pseudocode

```
START
STORE student tuples
SET seen IDs to empty list
FOR each record
    UNPACK ID, name, marks, attendance
    CHECK duplicate ID
    CHECK name is not blank
    CHECK marks and attendance are in 0..100
    DISPLAY status and errors
STOP
```

Python Source Code

```
records = [
    ("S01", "Asha", 85, 92),
    ("S02", "Bharat", 110, 80),
    ("S01", "Charan", 70, 75),
    ("S04", "", 65, 88)
]

seen = []

for sid, name, marks, attendance in records:
    errors = []

    if sid in seen:
        errors.append("Duplicate ID")
    else:
        seen.append(sid)

    if name.strip() == "":
        errors.append("Missing name")

    if marks < 0 or marks > 100:
        errors.append("Invalid marks")

    if attendance < 0 or attendance > 100:
        errors.append("Invalid attendance")

    if errors:
        print(sid, "-", ", ".join(errors))
    else:
        print(sid, "- Valid")
```

Test Cases

Test input / condition	Expected result
S01, Asha, 85, 92	Valid
S02 marks 110	Invalid marks
S01 used again	Duplicate ID
S04 blank name	Missing name

Output Screenshot

Sample console output (formatted to show the expected execution result):

```
S01 - Valid
S02 - Invalid marks
S01 - Duplicate ID
S04 - Missing name
```

Conclusion

The program checks consistency of student records using tuples, lists, loops and conditional statements.